Experiment No. 4
C++ INPUT/OUTPUT STATEMENTS

I. OBJECTIVES

1. To know the use of input and output statements.
2. To be able to differentiate and apply the input and output streams.

II. LABORATORY EXERCISES

Encode the following program and compile them. Illustrate the output.

Note: if getch(); will not work on other C++ ide compiler replace it with _getch();

Program no.1 (save as prog1.cpp)

```
#include <iostream>
#include<conio.h>
using namespace std;
int main ()
{
cout << "Hello, World! " ;
getch();
return 0;
}
```

Program No. 2 (save as prog2.cpp)

```
#include <iostream>
using namespace std;
int main ()
{
      int age;
      age = 16;
      cout << "My age is "<<age;
      cin.get();
      return 0;
}
```

Program No 3. (save as prog3.cpp)

```
#include <iostream>
#include<conio.h>
using namespace std;
int main ()
{
      char ch;
      ch = 'A';
      cout<<ch;
      ch = 'B';
      cout <<ch;
      getch();
      return 0;
}
```

Program No. 4(save as prog4.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
      int x, y;
      cout<<"enter value for  and y:";
      cin>>x>>y;
      cout<<"Quotient : "<<x/y<<"\n";
      cout<<"Remainder : "<<x%y;
      getch();
      return 0;
}
```

Program No. 5(save as prog5.cpp)

```
#include<iostream>
#include<conio.h>
#include<iomanip>
using namespace std;
int main()
{
// conversion feet to meters

      int feet;
      float meters;
      cout<<"Enter a number in feet : " ;
      cin>>feet ;
      meters=feet*0.3048;
      cout<<feet<<" feet is "<<setprecision(4)<<meters<<"meters";

      getch();
      return 0;
}
```

Program No.6 (save as prog6.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
      int i,j;
      cout<<"Enter two number: " ;
      cin >>i>>j;
      cout<<i<<"=="<<j<<"is"<<(i==j)<<"\n";
      cout<<i<<"!="<<j<<"is"<<(i!=j)<<"\n";
      cout<<i<<"<="<<j<<"is"<<(i<=j)<<"\n";
      cout<<i<<">="<<j<<"is"<<(i>=j)<<"\n";
      cout<<i<<"<"<<j<<"is"<<(i<j)<<"\n";
      cout<<i<<">"<<j<<"is"<<(i>j)<<"\n";
      getch();
      return 0;
```

```
}
```

Program No.7 (save as prog7.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
      int i, j;
      cout<<"Enter two number:" ;
      cin>>i>>j;
      cout<<i<<"AND"<<j<<"is"<<(i&&j)<<"\n";
      cout<<i<<"OR"<<j<<"is"<<(i||j)<<"\n";
      cout<<"NOT " <<j<<"is"<<(!i)<<"\n";
      getch();
      return 0;
}
```

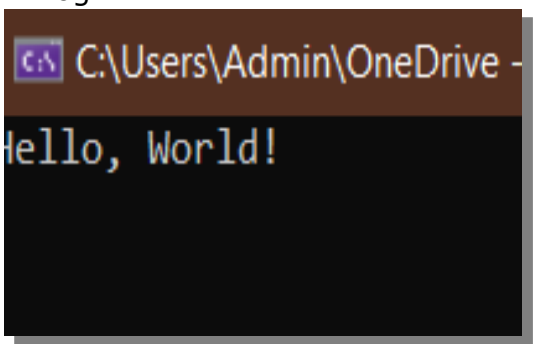
Program No.8(save as prog8.cpp)

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
      int a, b;
      a = b = 5;
      cout<<"\n"<<a--<<"  "<<--b;
      cout<<"\n"<<a--<<"  "<<--b;
      cout<<"\n"<<a--<<"  "<<--b;
      cout<<"\n"<<a--<<"  "<<--b;
      cout<<"\n"<<a--<<"  "<<--b;
      getch();
      return 0;
}
```

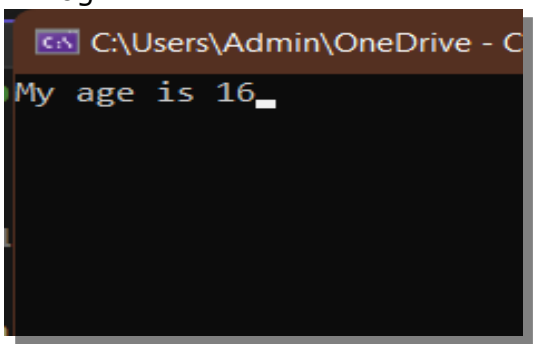
III. Summary of Program Outputs

Direction: Demonstrate the corresponding output for each of the given programs. Give your observation and analysis for each of the problems.

1. Prog1



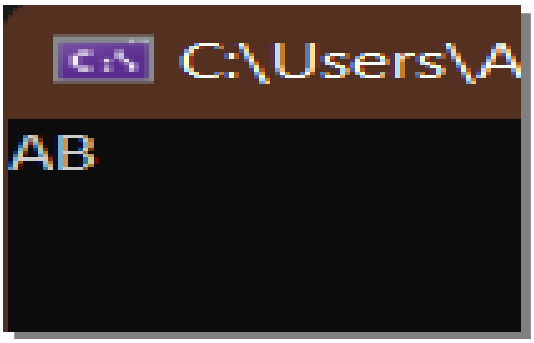
2. Prog2



Program No. 1 simply used the insertion operator "<<" and the "cout" object which refers to "Character Output," to display the text "Hello, World!".

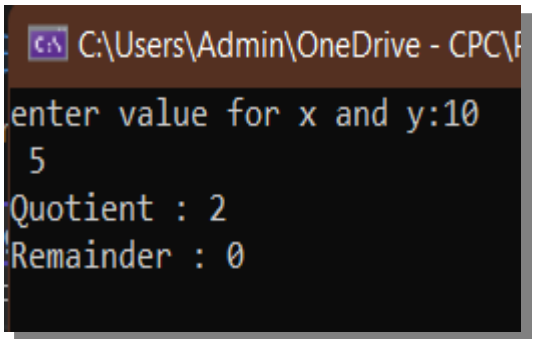
Program No. 2 used the data type “Int” which means “Integer” to store the value of the variable “age” which equals to16. Using the “Character Output” or “Cout” and insertion operator to display the word “My age is” and using another insertion operator to add the variable “age” which equals to16. So output will be “My age is 16”.

3 prog3



Program No. 3 declared one single character to variable "ch," which is "A," and displayed it using "Cout" plus the insertion operator plus the variable "ch." After that, it declares another single character to the variable, which is "B" and displays it using "cout" plus the insertion operator plus the variable "ch". And for the result, it will only display the two single characters that combine together, which is "AB".

4. Prog4



Program No. 4 used the object “Int” to declare a single value for the variable “X” and “Y”, and it display the word “enter a value for x and y” using “cout” plus the insertion operator. And to input, the value of the variables “X” and “Y” it used “cin” which means “Character Input” plus the extraction operator “>>”. And in order to display the quotient of the variable “X” and “Y”, it displays first the text “Quotient:” plus the formula to get the quotient which is “X/Y” and using “\n” to form a new line to display the “Remainder:” plus the formula which is “X%Y”. So the result will be, the program will ask the user to enter the value for X and Y and it will compute the quotient and Remainder to the inputted value of X and Y.

5 prog5

```
C:\Users\Admin\OneDrive - CPC\Pro
Enter a number in feet : 5
5 feet is 1.524meters
```

Program No. 5 Convert feet to meters by using the object "Int" to declare a single value in the variable "feet" and using "float" to declare a number that has a decimal value in the variable "meters". Using the object "Cout" and insertion operators to display the text "Enter a number in feet" and using the object "Cin" plus extraction operator plus the variable so that the user can input or store a value in the variable "Feet". In order to convert the feet to meters, it uses the formula "Meters = feet * 0.3048" and displays it using "cout" plus insertion operators plus the variable "feet" and inserts the text "feet is" plus the "setprecision" in order to use the formula to convert the feet to meters plus the variable "Meters" and the text "meters". So, the result will be, the program will ask the user to input a number in feet and convert the inputted number in feet to meters.

6. Prog6

```
C:\Users\Admin\OneDrive
Enter two number: 10 5
10==5is0
10!=5is1
10<=5is0
10>=5is1
10<5is0
10>5is1
```

Program No. 6 used “int” to declare a single value in the variable “i” and “j” and it also used the object “cin” to ask the user to input/stored a single value on the variable. Using “cout” to display the value of “i” plus insertion operator to insert the relation operator which is “==” plus the “j” variable plus another text “is” plus the expression or the decision “i == j” and end the line using “\n”. To sum it up, the program will ask the user to input a single value for the variable “I” and “j” and the program will evaluate the expression if it’s true or false if its true it will display “1” and if its false it will display "0".

7. Prog7

```
C:\Users\Admin\OneDrive
Enter two number:20 2
20AND2is1
20OR2is1
NOT 2is0
```

Program No. 7 used “int” to declare a value in the variable “I” and “J”. It used “cout” to display the text “Enter two numbers” and “cin” to input the value of the variable “I” and “J”. And it identifies if the two inputted number is true it will display “1” and if false it will display “0”.

8. Prog8

```
C:\Users\
5   4
4   3
3   2
2   1
1   0
```

Program No 8 declared a value of the variable “a” which is “b” with its own value of 5. This program will keep subtracting 1 from the value of the variable “b” using the decrement operator which is the “--“and displays it using the variable “a”.

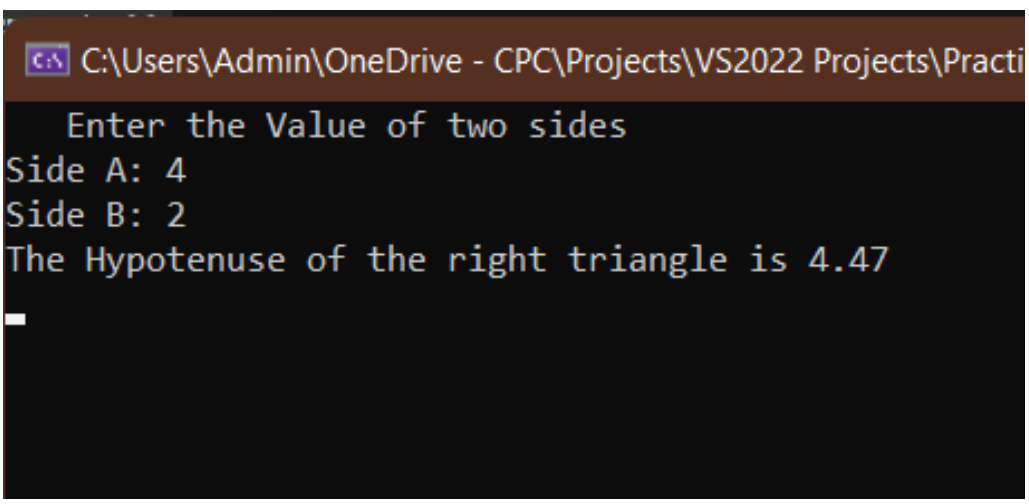
1. Write a program that will allow a user to find the hypotenuse of a right triangle using the Pythagorean theorem ($c^2 = a^2+b^2$). Round off to 2 decimal places.

```
#include <iostream>
#include <conio.h>
#include <math.h>
#include <iomanip>
using namespace std;


int main()
{
   int sideA, sideB;
   cout << "   Enter the Value of two sides \n";
   cout << "Side A: ";
   cin >> sideA;
   cout << "Side B: ";
   cin >> sideB;
   double C = sideA * sideA + sideB * sideB;
   C = sqrt(C);
   cout << "The Hypotenuse of the right triangle is " << setprecision(3) << C <<
endl;

          _getch();
          return 0;
}
```

Output:



2. Design a program to find the circumference of a circle. Use the formula: C=2∏r, where ∏ is approximately equivalent to 3.1416. Round off to 2 decimal places.

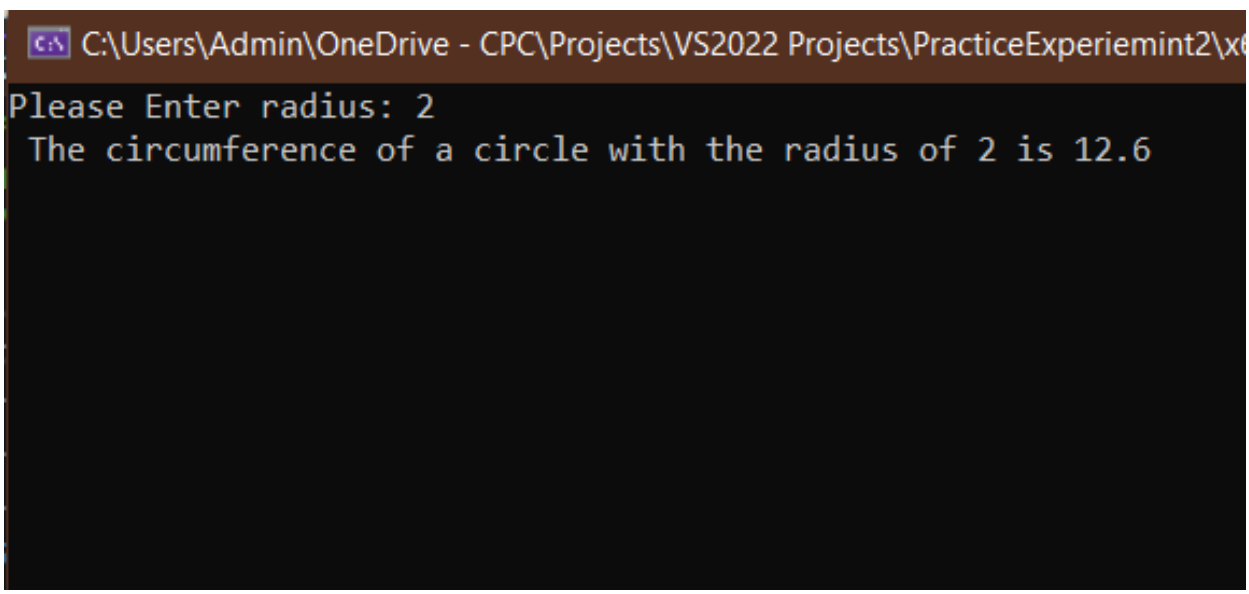
```
#include <iostream>
#include <conio.h>
#include <math.h>
#include <iomanip>
using namespace std;

int main()
{
    double pi = 3.1416;
    int rad;

    cout << "Please Enter radius: ";
    cin >> rad;
    double c = 2 * rad * pi;
    cout << " The circumference of a circle with the radius of " << rad << " is
" << setprecision(3) << c << endl;

    _getch();
    return 0;
}
```

Output:

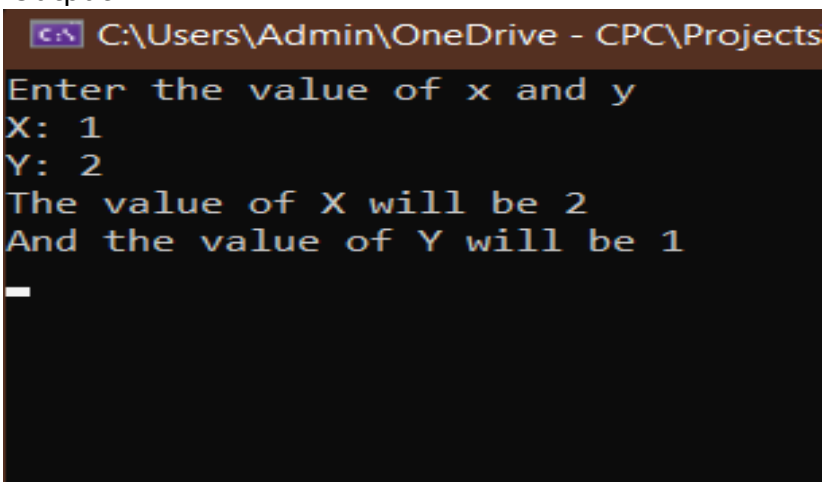


3. Write a program that exchanges the value of two variables: x and y. The output must be: the value of variable y will become the value of variable x, and vise versa.

```
#include <iostream>
#include <conio.h>
using namespace std;


int main()
{
      int x, y, z;
      cout << "Enter the value of x and y \n";
      cout << "X: ";
      cin >> x;
      cout << "Y: ";
      cin >> y;
    z = x;
      x = y;
      y = z;
      cout << "The value of X will be " << x << endl;
      cout << "And the value of Y will be " << y << endl;
      _getch();
      return 0;
}
```

Output:



V. Conclusion

I learned through this experiment that "using namespace std" is a group of related names or identifiers that serves to separate them from similar identifiers in other namespaces or the global namespace. Namespace std is where the C++ standard library's identifiers are stored. To handle input and output, you must first include the "conio.h" header file, after which you can make use of objects like "cin" to enter a value into a variable, "cout" to print or display text, and "_getch()" to pause the program's exit prompt until a key is pressed. I also learned several c++ data types including “int” which means integers that are usually used to indicate integers and “char” which means “character” that is used to declare character type variables.